SAAS CUSTOMER CHURN INTELLIGENCE

Business Analytics Consulting Engagement

Engagement Type: Customer Retention and Revenue Intelligence
Department: Data Analytics and Business Intelligence
Client Industry: Software as a Service (SaaS)
Assigned Role: Data Analyst, Customer Intelligence
Project Focus: Exploratory Data Analysis, SHAP Interpretation, and Business Recommendations

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1. Background and Business Context

A growing Software as a Service (SaaS) company has been experiencing customer churn, resulting in lost recurring revenue, reduced customer lifetime value, and uncertainty about the factors influencing customer retention.

Although the company collects data on customer payment behaviour, support interactions, product adoption, subscription plans, and revenue, management does not have a sufficiently clear understanding of how these factors relate to churn.

The company has commissioned your analytics consultancy to investigate the behaviour of churned businesses and identify actionable insights that can support better customer retention decisions.

As part of an earlier machine learning investigation, a Random Forest classification model was developed, and SHAP (SHapley Additive exPlanations) analysis was conducted to understand the features influencing its predictions. The model reportedly identified 98% of churned businesses under the evaluation metric used.

Your assignment is to conduct an independent, evidence-based exploratory analysis of the dataset, evaluate the insights generated by the model, and translate your findings into practical recommendations for business leaders.

Your responsibility is not just to describe what happened. You must explain what the data suggests, why the findings matter commercially, and what the company should do next.

2. Business Objectives

Your engagement must address the following objectives:

1. Investigate the characteristics and behaviour of churned businesses compared with retained businesses.
2. Identify patterns associated with payment status, support tickets, product adoption, subscription plans, and customer revenue.
3. Examine differences in churn behaviour across industries, countries, and customer managers.
4. Evaluate the business relevance of the SHAP insights generated from the Random Forest model.
5. Determine which customer segments may require retention interventions.
6. Translate analytical findings into recommendations that could reduce churn and protect monthly recurring revenue (MRR).

3. Scope of Investigation

Your analysis must cover the following business dimensions.

Business dimension	**Questions to investigate**
Payment status	Are churned businesses more likely to have failed, overdue, or incomplete payments?
Support tickets	Do churned businesses submit more support tickets? Are particular ticket volumes associated with higher churn?
Feature adoption rate	Are businesses with lower product adoption more likely to churn?
Total seats allowed	Does the permitted number of seats differ between churned and retained businesses?
Monthly recurring revenue (MRR)	How much recurring revenue is associated with churned businesses? Which customer segments contribute most to the loss?
Plan tier	Are churn rates different across subscription tiers?
Industry	Are particular industries disproportionately represented among churned businesses?

Country	Are there geographical differences in customer churn?
Customer manager	Do churn rates and churned MRR vary across managers?

You are expected to investigate these dimensions individually and in combination. For example, a low feature adoption rate may have a different business implication for an enterprise customer than for a small business on a basic subscription plan.

4. Required Analytical Tasks

Task 1: Data Understanding and Quality Assessment

Inspect the dataset before beginning the analysis.

Your responsibilities include:

- Identifying the number of rows and columns.
- Understanding the meaning and data types of each variable.
- Identifying missing values, duplicate records, inconsistent categories, and invalid values.
- Checking whether the churn variable is correctly represented.
- Examining the distributions of numerical and categorical variables.
- Identifying potential outliers in support tickets, feature adoption, seats, and MRR.
- Documenting data quality issues and the steps taken to address them.

Expected outcome: A clean, documented dataset that is suitable for exploratory analysis.

Task 2: Customer Churn Overview

Establish the current state of customer churn.

Investigate:

- Total number of businesses.
- Number and percentage of churned businesses.
- Number and percentage of retained businesses.
- Total MRR associated with churned businesses.
- Average and median MRR for churned and retained businesses.
- Distribution of churned businesses across subscription tiers.

Business question: How significant is the churn problem, and what is its potential financial impact?

Distinguish between the number of customers lost and the recurring revenue associated with those customers.

Task 3: Payment Behaviour and Churn

Investigate the relationship between payment status and churn.

Compare payment outcomes across churned and retained businesses. Determine whether failed, overdue, or other payment statuses are disproportionately associated with churn.

Examine whether payment behaviour differs by plan tier, industry, or customer manager.

Business questions:

- Are payment problems associated with higher churn rates?
- Which payment statuses appear most frequently among churned businesses?
- Could payment recovery processes help retain otherwise viable customers?

Do not automatically assume that payment failure caused churn. Establish the timing and meaning of the available payment data before drawing conclusions.

Task 4: Support Tickets and Customer Experience

Investigate how support ticket activity differs between churned and retained businesses.

Analyse:

- Average and median ticket volumes by churn status.
- Churn rate across support ticket volume groups.
- Ticket patterns by subscription tier and industry.
- Whether high ticket volumes coincide with low feature adoption or payment problems.

Business questions:

- Are churned businesses experiencing more support friction?
- Are there customer segments with unusually high support demand?
- Which patterns should customer success teams investigate further?

If ticket resolution time, ticket category, or customer satisfaction scores are available, incorporate them into the investigation.

Task 5: Feature Adoption and Product Engagement

Evaluate the relationship between feature adoption and customer retention.

Investigate:

- Feature adoption distributions for churned and retained businesses.
- Churn rates across low, medium, and high adoption groups.
- Adoption patterns by subscription tier and industry.
- The relationship between adoption and MRR.
- Whether customers with low adoption are concentrated in particular countries or under particular managers.

Business questions:

- Are customers using less of the product more likely to churn?
- Which customer segments have low adoption but significant recurring revenue?
- Where could onboarding, training, or product engagement interventions create value?

If adoption is recorded as a percentage, validate its range and definition before interpreting the results.

Task 6: Seats, Subscription Plans, and Revenue

Examine the relationship between customer size, subscription tier, MRR, and churn.

Investigate:

- Differences in permitted seats between churned and retained businesses.
- Churn rates by plan tier.
- Average MRR by plan tier and churn status.
- The distribution of churned MRR across customer segments.
- Whether low adoption is associated with underutilised seats, where actual seat usage is available.

Business questions:

- Which subscription tiers have the highest churn rates?
- Which tiers account for the greatest amount of churned MRR?
- Are some high-value customers leaving despite having substantial subscription allowances?
- Are there opportunities to improve customer fit, onboarding, or plan utilisation?

Do not assume that a high seat allowance means a customer is actively using those seats. If actual seat utilisation is unavailable, state that limitation.

Task 7: Industry, Country, and Customer Manager Analysis

Examine whether churn patterns differ across industries, countries, and customer managers.

For each dimension:

- Calculate the number of businesses and churn rate.
- Calculate churned MRR.
- Compare churned and retained customer characteristics.
- Identify segments that warrant further investigation.
- Examine whether other variables, such as plan tier or adoption, may help explain observed differences.

For customer managers, use a fair comparison that considers portfolio size and customer mix. A manager with more churned customers may also manage substantially more customers overall.

Business questions:

- Which industries and countries have the highest churn rates?
- Which segments account for the greatest financial exposure?
- Are some managers associated with stronger retention outcomes after considering portfolio composition?
- What additional evidence would be needed before recommending changes to customer management practices?

Protect customer and employee confidentiality when presenting results.

Task 8: SHAP Insight Evaluation

The consultancy has access to SHAP outputs generated from a Random Forest model that reportedly identifies 98% of churned businesses.

Your responsibility is to investigate whether the model's explanations align with the patterns observed in the dataset.

Review the available SHAP feature importance ranking, SHAP summary or beeswarm plot, and individual waterfall plots where available.

Your analysis must address:

- Which features contribute most to the model's churn predictions?
- Which feature values are associated with higher or lower predicted churn risk?
- Do the most influential features show similar patterns in your EDA?
- Are the findings consistent across customer segments?
- Which insights could support practical retention decisions?
- Which explanations require additional investigation before management acts on them?

For example, if SHAP suggests that low feature adoption contributes strongly to predicted churn, compare adoption distributions and churn rates across adoption groups. If payment status is influential, examine the corresponding payment patterns.

Important: SHAP explains how features contribute to a model's predictions. It does not, by itself, establish that a feature causes churn. Your report must distinguish model explanations, observed associations, and causal conclusions.

Also verify what the reported 98% means. If it is recall, the model identifies 98% of actual churners in the evaluated sample. If it is accuracy, it means 98% of all evaluated predictions were correct. These are not equivalent.

Where possible, report recall, precision, F1 score, ROC-AUC, and the confusion matrix, using an appropriate held-out test set. Check class imbalance and the business cost of false positives and false negatives.

If you only receive SHAP plots rather than the model, prediction data, or evaluation results, clearly state that your assessment is limited to the supplied evidence.

Task 9: Develop Business Recommendations

Translate your findings into specific actions for the SaaS company's management team.

Your recommendations should address:

- Payment recovery and billing intervention.
- Customer onboarding and feature adoption.
- Support quality and customer experience.
- High-value customer retention.
- Industry or country-specific retention strategies.
- Customer manager portfolio improvement.
- Risk-based customer prioritisation using model outputs.

Each recommendation must be linked to an observed finding or a validated model insight.

For every major recommendation, explain the evidence, the customer segment affected, the proposed action, the expected business benefit, and the KPI management should monitor.

Where the data does not establish an outcome, describe the expected benefit as a hypothesis to test rather than a guaranteed result.

5. Required Deliverables

Submit the following consulting outputs:

Deliverable 1: Data Quality and Preparation Report

Document the initial data assessment, cleaning decisions, assumptions, and limitations.

Deliverable 2: Exploratory Data Analysis Notebook

Provide a reproducible Python notebook containing the analysis, calculations, visualisations, and interpretations.

Deliverable 3: Executive Churn Intelligence Dashboard

Develop a Power BI dashboard that allows management to explore churn rates, churned MRR, payment behaviour, adoption, support tickets, subscription tiers, industries, countries, and customer manager portfolios.

Deliverable 4: SHAP Insight Evaluation

Provide a concise assessment comparing the model's leading explanations with observed data patterns. Include appropriate plots or references to supplied SHAP outputs.

Deliverable 5: Executive Consulting Report

Prepare a business-focused report summarising the major findings, financial implications, priority customer segments, limitations, and recommendations.

Deliverable 6: Management Presentation

Deliver a concise presentation explaining what management needs to know, why it matters, and what actions should be prioritised.

6. Expected Analytical Standards

Your work will be assessed on the following principles:

1. **Accuracy:** Calculations and interpretations must be correct.
2. **Analytical depth:** Go beyond totals and basic charts to investigate meaningful patterns.
3. **Business relevance:** Connect findings to retention, recurring revenue, and customer success.
4. **SHAP interpretation:** Explain model outputs correctly and compare them with observed data.
5. **Visual communication:** Build clear, appropriate, and well-labelled charts and dashboards.
6. **Evidence-based recommendations:** Every major recommendation must be supported by evidence.
7. **Professional communication:** Present your conclusions in language that a non-technical executive can understand.
8. **Reproducibility and integrity:** Document assumptions, avoid unsupported claims, and make your analysis repeatable.

7. Final Management Questions

At the conclusion of the engagement, your consultancy must be able to answer these questions:

1. How serious is the company's churn problem?

2. Which customer segments account for the greatest number of churned businesses?
3. Which segments account for the greatest amount of lost recurring revenue?
4. What payment, support, and product engagement patterns distinguish churned businesses?
5. Which industries, countries, plans, and manager portfolios deserve further investigation?
6. Do the SHAP explanations align with the patterns observed in the data?
7. What are the three highest-priority retention interventions?
8. How should management measure whether those interventions work?

8. Definition of Success

The engagement will be considered successful when management receives a clear, evidence-based understanding of churn patterns, a defensible evaluation of the supplied SHAP insights, and a prioritised set of actions that can be tested and measured.

The ultimate objective is not to produce the largest number of charts. It is to turn customer data into better retention decisions and protect recurring revenue.

Your role is to think, investigate, challenge assumptions, and communicate like a professional analytics consultant.